

- ① A galvanometer of resistance 50Ω is converted into a voltmeter of range $(0-2V)$ using a resistor of $1K\Omega$. If it is to be converted into a voltmeter of range $(0-10V)$, the resistance required will be.
A) $4.8K\Omega$ B) $5K\Omega$ C) $5.2K\Omega$ D) $5.4K\Omega$
- ② A pure Si Crystal having 5×10^{28} atoms/ m^3 is doped with 1ppm concentration of Antimony. If the concentration of holes in the doped crystal is found to be $4.5 \times 10^9 / m^3$, the concentration in m^{-3} of intrinsic charge carriers in Si crystal is about.
A) 1.2×10^{15} B) 1.5×10^{16} C) 3×10^{15} D) 2×10^{16}
- ③ The potential energy of two nucleons (between them) inside a nucleus is minimum at a distance of about.
A) $0.8 fm$ B) $1.6 fm$ C) $2.0 fm$ D) $2.8 fm$
- ④ A telescope has an objective lens of focal length $150cm$ and an eye piece of focal length $5cm$. Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective.
- ⑤ a) Two energy levels of an electron in hydrogen atom are separated by $2.55 eV$. Find the wavelength of radiation emitted when the electron makes transition from the higher energy level to the lower one.
b) In which series of hydrogen spectrum this line shall fall?